ECE 501 Project Week 5 Report

Project Title: Vehicle Orientation Estimation Using Conventional Image Processing

Group Number: 14

Project number: 8

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Github Repository Link: https://github.com/hetpatel5107/DIP_2026_Project8_Group14

- **Introduction**

In Week 5, we explored the Radon transform as another conventional image processing method for estimating vehicle orientation. In the previous weeks, we used the Sobel method to estimate orientation from the dominant gradient direction. However, Sobel can be affected by edges from vehicle structures, shadows, and background objects.

The Radon transform analyses the overall line structure of an image by calculating projections at different angles. This week, we applied Canny edge detection followed by the Radon transform to the same cropped Sedan image used in Week 3. We estimated its main axis orientation and compared the result with the ground-truth orientation and the previous Sobel estimate.

- **Canny edge detection method**

Code:

```
edges = canny(gray.astype(float) / 255.0, sigma=1.0)



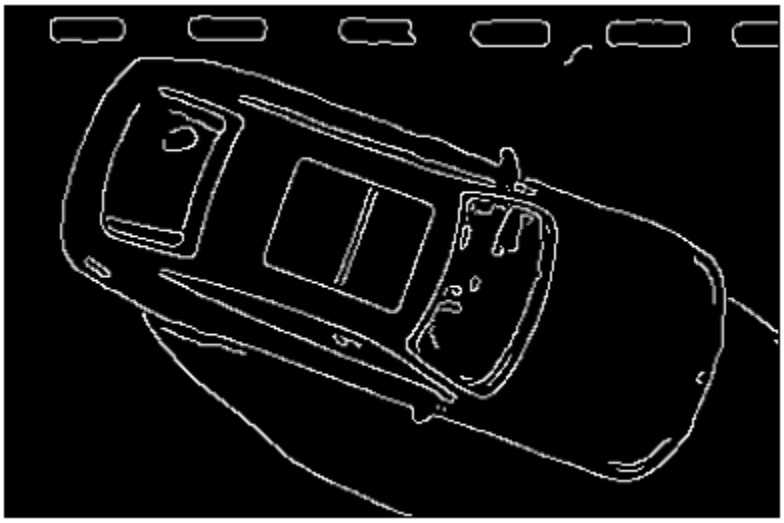
plt.figure(figsize=(5, 5))

plt.imshow(edges, cmap="gray")

plt.axis("off")

plt.show()
```

Output:



What it is & Why we use it

Canny is a tool we use to find the clean, structural outline of the car. It smooths out random camera static, finds the sharpest changes in pixel brightness, and thins those

edges down to a single-pixel line. This gives us a crisp, black-and-white edge map of the vehicle.

Why pair it with Radon?

If we fed the raw photo straight into the Radon transform, it would get completely confused by shadows, windshield glare, and the car's paint job. Running Canny first reduces the influence of broad intensity variations and provides an edge map for Radon to analyse. This gives Radon perfectly clean lines to analyze. However, edges from shadows and background structures may still remain.

Mathematical Expression:

Canny uses two thresholds (T_{high} and T_{low}) to filter the edge gradients (G):

- If $G \geq T_{high}$, we keep it as a strong edge.
- If $T_{low} \leq G < T_{high}$, it's a weak edge.
- If $G < T_{low}$, we throw it out as noise.

Explanation:

This snippet just takes our cropped grayscale car, scales the pixels, and applies the Canny filter to pull out the boundaries. It spits out a clean binary image where the white pixels are the car's outline and the black pixels are just empty space.

- **Radon method**

The Radon transform represents an image by calculating sums of its pixel values along straight lines at different angles and positions. It therefore measures how image or edge information is distributed along lines throughout the image. Repeating the projection for many angles produces a sinogram, where one axis represents projection angle and the other represents line position.

A line can be described by r = x cos(θ) + y sin(θ), where θ is the angle of the line's normal and r is its perpendicular distance from the origin. The Radon transform R(r, θ) is the integral of the image along that line. In a digital image, the operation is approximated using pixel values.

$$R(r, \theta) = \iint E(x,y)\, \delta(r - x\cos(\theta) - y\sin(\theta))\, dx\, dy$$

Sobel estimates local intensity-change directions, so its dominant response may be influenced by windows, shadows, roof edges, and other structures that do not represent the vehicle's main axis. We therefore tested Radon, which aggregates edge responses along lines over the whole crop. In this implementation, the angle is selected using the variance of the projection values:

$$\theta^* = \arg\max_\theta \mathrm{Var}_r[R(r, \theta)]$$

Code:

```
theta = np.arange(0, 180, 1)



sinogram = radon(

    edges.astype(float),

    theta=theta,

    circle=False

)



plt.figure(figsize=(10, 5))

plt.imshow(

    sinogram,

    cmap="gray",

    aspect="auto",

    extent=[theta[0], theta[-1], sinogram.shape[0], 0]

)
```

```
plt.xlabel("Projection Angle (degrees)")

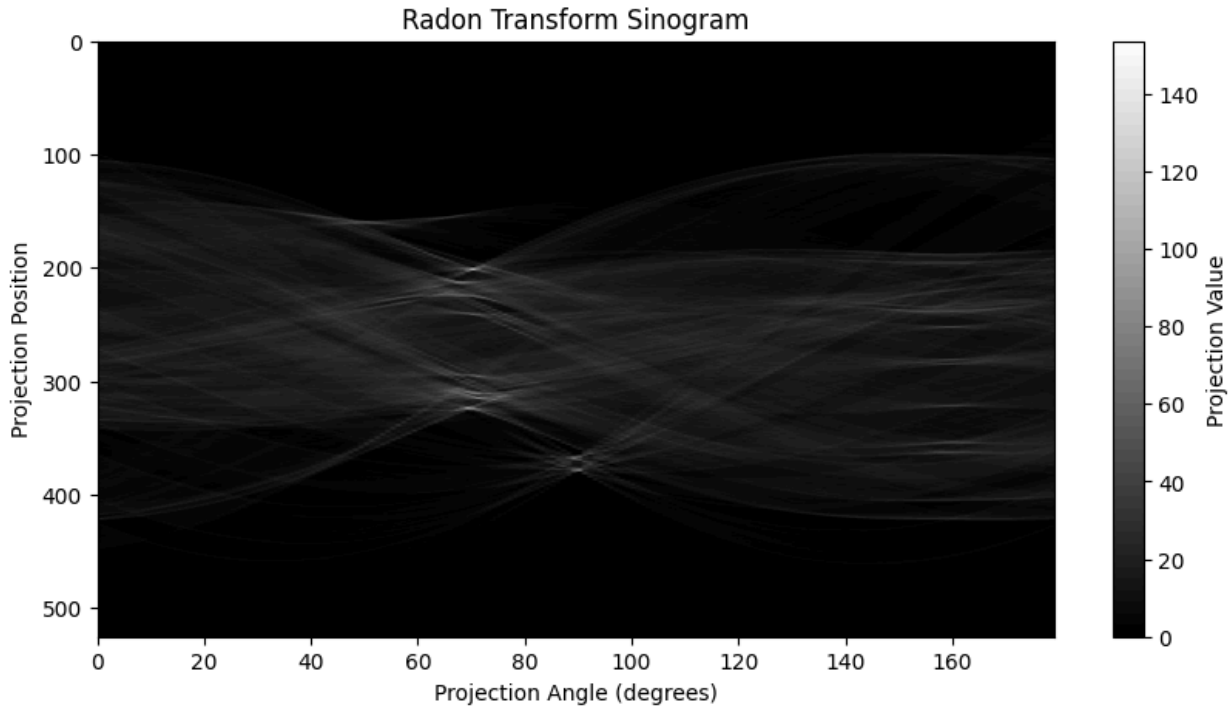
plt.ylabel("Projection Position")

plt.title("Radon Transform Sinogram")

plt.colorbar(label="Projection Value")

plt.show()
```

Output:



Explanation:

The Radon projection angle describes a line's normal, so the corresponding line axis angle is obtained by adding 90° and taking the result modulo 180°. The resulting axis is undirected and cannot distinguish the front of a vehicle from its rear.

- **Results and Comparison**

1. Estimation of the Vehicle's Main Axis Orientation

Code :

```
projection_variance = np.var(sinogram, axis=0)



best_projection_angle = theta[np.argmax(projection_variance)]



radon_axis_angle = (best_projection_angle + 90) % 180

def convert_to_0_90(angle):

    angle = angle % 180

    if angle > 90:

        angle = 180 - angle

    return angle



radon_estimate = convert_to_0_90(radon_axis_angle)

ground_truth_converted = convert_to_0_90(ground_truth)



print("Best Radon Projection Angle:", best_projection_angle, "degrees")

print("Radon Estimated Axis:", round(radon_estimate, 2), "degrees")

print("Ground Truth Axis:", round(ground_truth_converted, 2), "degrees")
```

Output:

```
Best Radon Projection Angle: 70 degrees
Radon Estimated Axis: 20 degrees
Ground Truth Axis: 17.1 degrees
```

Result and Observation:

The Radon transform estimated the vehicle's main axis orientation as 20°, while the ground-truth orientation was 17.1°. The absolute difference between the estimated and ground-truth orientations was 2.9°. This indicates that the Radon based method produced an estimate close to the ground-truth orientation for the selected sedan image.

2. Comparison of Sobel and Radon Estimates with Ground Truth

Code:

```
radon_error = abs(radon_estimate - ground_truth_converted)

sobel_error = abs(sobel_estimate - ground_truth_converted)



print("Ground Truth Angle:", round(ground_truth_converted, 2))

print("Sobel Estimated Angle:", round(sobel_estimate, 2))

print("Sobel Error:", round(sobel_error, 2))

print("Radon Estimated Angle:", round(radon_estimate, 2))

print("Radon Error:", round(radon_error, 2))
```

Output:

```
Ground Truth Angle: 17.1
Sobel Estimated Angle: 11.0
Sobel Error: 6.1
Radon Estimated Angle: 20
Radon Error: 2.9
```

Result and Observation:

The Sobel method produced an orientation error of 6.1°, while the Radon method produced an error of 2.9°. For this particular sedan image, the Radon method had a 3.2° lower error than the Sobel method.

The absolute orientation error is calculated as the difference between the estimated angle and the ground-truth angle, ignoring the sign.

For Sobel:

Sobel Error = |11.0 − 17.1| = 6.1°

For Radon:

Radon Error = |20.0 − 17.1| = 2.9°

These results suggest that the Radon method performed better for this particular image. However, additional vehicle crops must be tested before concluding that Radon is generally more accurate than Sobel.

- **Conclusion**

This week, we implemented a Radon-based approach using Canny edge detection to estimate the main axis orientation of a cropped vehicle image. The estimated orientation was compared with the ground truth and the existing Sobel estimate.

For the selected Sedan crop, Radon achieved an error of 2.9°, lower than the Sobel error of 6.1°. To see if the Radon method is actually better than Sobel, we will have to test it on multiple cropped images.

1. Work done this week

- Applied Canny edge detection to a cropped vehicle image.
- Applied the Radon transform across multiple projection angles.
- Displayed the resulting sinogram.
- Used projection variance to select the best projection angle.
- Converted the projection angle into an estimated vehicle-axis orientation.
- Compared the Radon estimate with ground truth and the existing Sobel estimate.
- Calculated the orientation error for both methods.

2. Work to be done next week

- Test the Radon-based method on additional vehicle crops.
- Compare the estimated orientations with the ground-truth annotations.
- Evaluate the errors across multiple vehicles.
- Identify the situations in which Radon performs well or produces large errors.
- Investigate possible improvements to the orientation-estimation method.

3. Challenges faced

- Understanding the difference between the Radon projection angle and the corresponding line-axis angle.
- Converting the estimated and ground-truth angles into a common 0°–90° range.
- Selecting the principal orientation from the Radon projection values.